

University Interscholastic League
2022 – 2023 Elementary Number Sense Test

A

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

Read Directions Carefully
Before Beginning Test

Do Not Unfold This Sheet
Until Told to Begin

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $34 - 23 =$ _____</p> <p>(2) $8 \times 9 =$ _____</p> <p>(3) $223 + 222 =$ _____</p> <p>(4) $2022 \div 2 =$ _____</p> <p>(5) $3 \times 5 \times 4 =$ _____</p> <p>(6) $132 \div 11 =$ _____</p> <p>(7) $11 + 12 + 13 =$ _____</p> <p>(8) $458 - 205 =$ _____</p> <p>(9) $32 \times 5 =$ _____</p> <p>*(10) $2296 + 1892 =$ _____</p> <p>(11) $11 \times 34 =$ _____</p> <p>(12) Which digit is in the thousandths place in 94371.65028? _____</p> <p>(13) $32 \times 12 =$ _____</p> <p>(14) 70836.4728 rounded to the thousandths place is _____ (decimal)</p> <p>(15) What is the remainder for $71532 \div 4$? _____</p> <p>(16) The number of odd whole numbers between 9 and 25 is _____</p> <p>(17) $8 \times 10^3 + 4 \times 10^1 + 4 \times 10^{-1} =$ _____ (decimal)</p> <p>(18) $18 \times 6 + 18 \times 4 =$ _____</p> <p>(19) MMDCL = _____ (Arabic Numeral)</p> | <p>*(20) $149 \times 319 =$ _____</p> <p>(21) $12 - 4 \div 2 =$ _____</p> <p>(22) $9 + 12 + 15 + 18 + 21 =$ _____</p> <p>(23) 2 days = _____ hours</p> <p>(24) $\frac{3}{4} =$ _____ percent</p> <p>(25) $\frac{5}{18} - \frac{1}{18} =$ _____</p> <p>(26) $88 \times 91 =$ _____</p> <p>(27) 0.55 = _____ common fraction</p> <p>(28) If 18 ♥ costs 43¢ then 36 ♥ cost _____ ¢</p> <p>(29) $\frac{5}{12} \times 36 =$ _____</p> <p>*(30) $111 \times 359 =$ _____</p> <p>(31) 30 nickels = _____ quarters</p> <p>(32) The product of the two largest primes less than 20 is _____</p> <p>(33) 3 cups = _____ ounces</p> <p>(34) $1225 \div 25 =$ _____</p> <p>(35) $44\frac{4}{9}\% =$ _____ common fraction</p> <p>(36) The LCM of 12 and 18 is _____</p> <p>(37) $103 \times 104 =$ _____</p> |
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- (38) $2\frac{4}{9} \times 7\frac{4}{9} =$ _____ (mixed number)
- (39) The ratio in inches of 2 yards to 1 foot is _____
- *(40) $167 \times 603 =$ _____
- (41) $17^2 =$ _____
- (42) $5^3 =$ _____
- (43) The side for a cube with volume 64 cm^3 is _____ cm
- (44) The perimeter of a square with area 361 m^2 is _____ m
- (45) $5\frac{3}{5} \div \frac{1}{5} =$ _____
- (46) $\sqrt{576} =$ _____
- (47) What is the perimeter of a right triangle with legs 3 and 4? _____
- (48) $22 \times 13 =$ _____
- (49) If $x = 15$, then $21 + 3x =$ _____
- *(50) $13 + 2023 + 2975 =$ _____
- (51) What is the number, **k**, in the sequence:
1, 4, 9, 25, **k**, 49, 64, . . .? _____
- (52) $12\frac{4}{9} - 7\frac{7}{9} =$ _____ (mixed number)
- (53) If the circumference of a circle is 144π , what is the radius of the circle? _____
- (54) What is the volume of a rectangular box that measures 5" by 6" by 8"? _____ in^3
- (55) 23 (base 4) = _____ (base 10)
- (56) What whole number cubed plus eight equals thirty-five? _____
- (57) A triangle has sides of 10, 12, and semiperimeter of 18. What is the third side? _____
- (58) If set $A = \{2, 4, 6, \dots, 10\}$ and set $B = \{1, 2, 3, 4, 5\}$, then the number of elements in $A \cup B$ is _____
- (59) What is the perimeter of the square with a side length of $8\frac{3}{4}$? _____
- *(60) 361 days = _____ hours
- (61) 57 (base 8) = _____ (base 2)
- (62) $-2^4 \times (-16) =$ _____
- (63) Two fair dice are thrown. What is the probability that the sum of the two sides showing is 9? _____
- (64) 1 square mile = _____ acres
- (65) $43^2 =$ _____
- (66) $4^6 \div 7$ has remainder of _____
- (67) How many edges does a square pyramid have? _____
- (68) If $2x + 12 < 8$, then $x <$ _____
- (69) $\frac{6}{7} + \frac{7}{6} = 2 +$ _____
- *(70) $102 \times \sqrt{784} =$ _____
- (71) The multiplicative inverse of $-3\frac{1}{2}$ is _____
- (72) The area of a square with diagonal 12 is _____
- (73) If $13\frac{1}{2}\%$ of x is $4\frac{1}{2}\%$ of 6, then $x =$ _____
- (74) $(-24) \div (-6) \times (-2) =$ _____
- (75) $24^2 + 12^2 =$ _____
- (76) $\left(25\frac{1}{2}\right)^2 - \left(14\frac{1}{2}\right)^2 =$ _____
- (77) What is the area of a trapezoid with bases 12, 16 and height 10? _____
- (78) $286 \times 28 =$ _____
- (79) $19^2 + 19 =$ _____
- *(80) $39 \times 40 \times 42 =$ _____

2022 – 2023 University Interscholastic League Elementary Number Sense Test A – Key

(1) 11	*(20) 45155 – 49907	(38) $18\frac{16}{81}$	(59) 35
(2) 72	(21) 10		*(60) 8231 – 9097
(3) 445	(22) 75	(39) 6	(61) 101111
(4) 1011	(23) 48	*(40) 95666 – 105736	(62) 256
(5) 60	(24) 75	(41) 289	(63) $\frac{1}{9}$
(6) 12	(25) $\frac{2}{9}$	(42) 125	(64) 640
(7) 36		(43) 4	(65) 1849
(8) 253	(26) 8008	(44) 76	(66) 1
(9) 160	(27) $\frac{11}{20}$	(45) 28	(67) 8
*(10) 3979 – 4397		(46) 24	(68) -2
(11) 374	(28) 86	(47) 12	
(12) 0	(29) 15	(48) 286	(69) $\frac{1}{42}$
(13) 384	*(30) 37857 – 41841	(49) 66	*(70) 2714 – 2998
(14) 70836.473	(31) $5\frac{3}{5}; \frac{18}{5}; 5.6$	*(50) 4761 – 5261	
(15) 0	(32) 323	(51) 36	(71) $-\frac{2}{7}$
(16) 7	(33) 24	(52) $4\frac{2}{3}$	(72) 72
(17) 8040.4	(34) 49	(53) 72	(73) 2
(18) 180		(54) 240	(74) -8
(19) 2650	(35) $\frac{4}{9}$	(55) 11	(75) 720
	(36) 36	(56) 3	(76) 440
	(37) 10712	(57) 14	(77) 140
		(58) 8	(78) 8008
			(79) 380
			*(80) 62244 – 68796

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.